

CAT 2025

MBA Fastrack



Lecture - 5

ALGEBRA

Inequalities

By- RAHUL BATHLA



TOPICS

to be covered

- 
- 
- 1 Linear Inequalities
 - 2 Quadratic Inequalities
 - 3 Special Cases



Topics : Linear Inequality

A Equations

$$\frac{3}{2}x + 5 = 14$$

$$\frac{3}{2}x = 9$$

$$x = 6$$

point solⁿ



$>$ $\geq$
 $<$ $\leq$

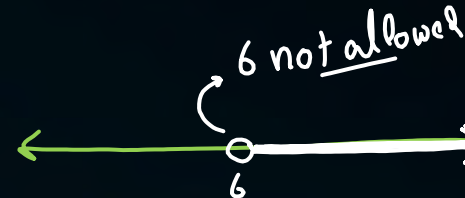
B Inequalities



$$\frac{3}{2}x + 5 > 14$$

$$\frac{3}{2}x > 9$$

RANGE OF SOLⁿ



$$x > 6$$

$$x \in (6, \infty)$$

open bracket

$$3x - 4 > 5$$

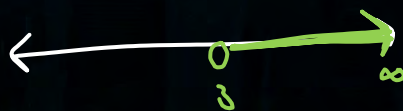
$$3x > 9$$

$$x > 3$$

open

$$x \in (3, \infty)$$

3 not inc



close

$$x \in [3, \infty)$$

3 is inc

$$3x - 4 \geq 5$$

$$3x \geq 9$$

$$x \geq 3$$





3 Golden Rules



Golden Rule I

- Never Cancel the negative sign on both the sides of inequalities.

$$\begin{array}{l} -x = -3 \\ x = 3 \end{array} \quad \checkmark$$

$$\begin{array}{l} -x > -3 \\ x < 3 \end{array} \quad \checkmark$$

$$\begin{array}{l} -4 > -10 \\ 4 < 10 \end{array} \quad \checkmark$$

- In case you do, Change the inequality sign.

$$\boxed{-x \geq 3}$$

$$\boxed{x \leq -3}$$

$$\underbrace{x - 3}_{> 0}$$

$$x > 3$$



Golden Rule 2



- Never Cancel the variables on both the sides.

if $x \neq 0$

$$x^x = x \quad \checkmark$$
$$x = 1$$

$$(-2)^2 > -2$$
$$-2 > 1 \quad \times$$

If the variable is **positive**
(cancel them)

$$x^2 \geq x$$
$$x \geq 1$$

If the variable is **negative**
(cancel them & change
inequality sign)

$$x^2 \geq x$$
$$x \leq 1$$

If the variable is **unknown**
(don't cancel)

$$x^2 \geq x$$
$$x^2 - x \geq 0$$
$$x(x-1) \geq 0$$



Golden Rule 3



- Never cross multiply in case of inequalities directly.

$$\frac{-2}{-1} > 1 \quad \times$$
$$-2 > -1$$

If the denominator is **positive**
(cross multiply it)

$$\frac{x}{y} > 2 \quad y \rightarrow +ve$$
$$x > 2y$$

If the denominator is **negative**
(cross multiply & change
inequality sign)

$$\frac{x}{y} > 2 \quad y \rightarrow -ve$$
$$x < 2y$$

If the denominator is **unknown**
(don't cross multiply)

$$\frac{x}{y} > 2$$
$$\frac{x-2y}{y} > 0$$
$$\frac{x}{y} - 2 > 0$$

QUESTION-1



#Q. $-3x - 5 < x + 7$

Find the Solution set of x ?

$$-3x - x < 7 + 5$$

$$-4x < 12$$

$$x > \frac{12}{-4}$$

$$x > -3$$

$$(-3, \infty)$$

$$-3 < x$$

Find the value of x if

- If x is Negative Integer.

$$-2 \neq -1$$

- If x is Positive Integer.

$$1, 2, 3, 4, \dots$$

- If X is non positive Integer.

$$-2, -1 \neq 0$$

- Represent the value of x on number line ?



QUESTION- 2



#Q. $5 - 2x \leq \frac{x}{6} - 8$. Find the Solution set of x ?

- A** $x > 6$
- B** $x < 6$
- C** $x \geq 6$ ✓✓
- D** $x \leq 6$

$$\cancel{13} \leq \frac{\cancel{13}x}{6}$$

$$6 \leq x$$



QUESTION- 3



#Q. The marks obtained by a student of Class XI in first and second terminal examination are 62 and 48, respectively. Find the minimum marks he should get in the third examination to have an average of at least 60 Marks in all 3 combinedly.

A 60

☒ **B** 70

C 80

D 90

$$\frac{62 + 48 + x}{3} \geq 60$$

$$x \geq 70$$

QUESTION- 4



#Q. $-11 < 3 + 4x \leq 9 - 2x$.

Find the Solution set of x ?

$$-11 < 3 + 4x$$

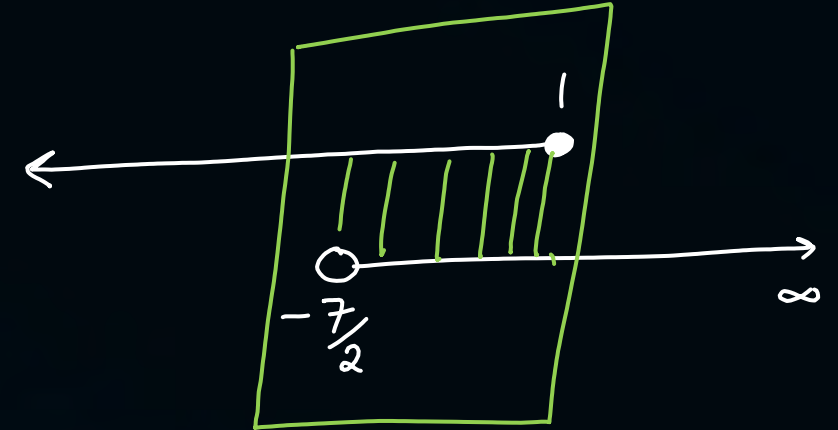
$$-14 < 4x$$

$$-\frac{7}{2} < x$$

$$3 + 4x \leq 9 - 2x$$

$$6x \leq 6$$

$$x \leq 1$$



$$x \in \left(-\frac{7}{2}, 1\right]$$



Topics : Quadratic Inequality

Quadratic Inequality Steps



$$\bullet \quad x^2 - 8x - 20 > 0$$

$$x^2 - 10x + 2x - 20$$

$$x(x - 10) + 2(x - 10)$$

$$(x - 10)(x + 2)$$

STEP 1 Make RHS 0 and FACTORISE

$$(x - 10)(x + 2) > 0$$

$$x - 10 = 0$$

$$x = 10$$

$$x + 2 = 0$$

$$x = -2$$

STEP 2 Draw the number line and mark Critical points.



$$x = 0$$

$$(0 - 10)(0 + 2)$$

$$(-)(+)$$

$$(-ve)$$

$$x^2 - 8x - 20$$

$$(x - 10)(x + 2)$$

$$x = 12$$

$$(12 - 10)(12 + 2)$$

$$(+) (+)$$

$$(+ve)$$

STEP 3 Mark the Signs of expression in each region



$$(x - 10)(x + 2) > 0$$

$$+ve$$

$$(x - 10)(x + 2) > 0$$

STEP 4 Choose your required region

$$(-\infty, -2) \cup (10, \infty)$$

$$(-\infty, -2] \cup [10, \infty)$$

Making the Wavy Curve - Shortcut



$$x^2 - 9x + 20 \leq 0$$

$$(x-4)(x-5) \leq 0$$

$\begin{matrix} -ve \\ \times \\ 0 \end{matrix}$



$$x \in [4, 5]$$

~~$$(x-3)(7+x)(9-x) \leq 0$$~~

$$(x-3)(x+7)(x-9) \geq 0$$

~~$$(x-2)(7-x)(2x-8) \leq 0$$~~

$$(x-2)(x-7)(2x-8) \geq 0$$

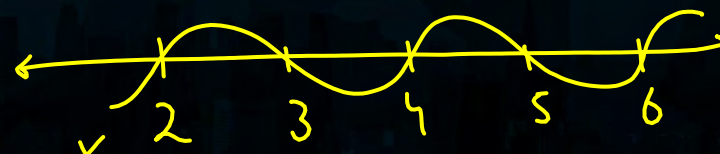
$$x \neq 0$$



$$[2, 4] \cup [7, \infty)$$

$$(x-2)(3-x)(x-4)(5-x)(x-6) < 0$$

$$(x-2)(x-3)(x-4)(x-5)(x-6) < 0$$



$$(-\infty, 2) \cup (3, 4) \cup (5, 6)$$

QUESTION- 5



#Q. $x^2 - 11x + 18 \leq 0$

- A** $[2, 9]$ ✓
- B** $(-\infty, 2] \cup [9, \infty)$
- C** $(-\infty, 2) \cup (9, \infty)$
- D** $(2, 9)$

$$(x-2)(x-9) \leq 0$$



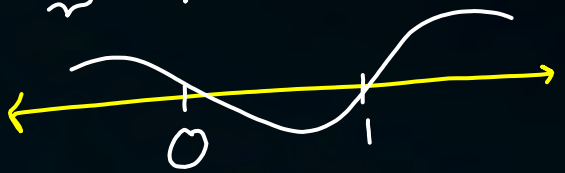
$$[2, 9]$$

QUESTION- 6



#Q. $x^2 > x$

- A** $(1, \infty)$
- B** $(-\infty, 0) \cup [1, \infty)$
- C** $(-\infty, 0) \cup (1, \infty)$ ✓✓
- D** $[1, \infty)$

$$x^2 - x > 0$$
$$\underbrace{(x)}_{< 0} \underbrace{(x-1)}_{> 0} > 0$$

$$(-\infty, 0) \cup (1, \infty)$$

QUESTION- 7



#Q. Find the range of x for which $(x - 4)(x - 5) > 6$?

A $(-\infty, 2) \cup (7, \infty)$ ✓

B $(-\infty, -2) \cup (7, \infty)$

C $(-\infty, -2) \cup (10, \infty)$

D $(-\infty, -2) \cup (0, \infty)$

$$\begin{cases} x^2 - 9x + 20 - 6 > 0 \\ x^2 - 9x + 14 > 0 \\ (x - 2)(x - 7) > 0 \end{cases}$$

#Q. For how many integers n , will the given inequality be satisfied ?

$$(n - 5)(n - 10) - 3(n - 2) \leq 0$$

$$n^2 - 18n + 56 \leq 0$$

$$(n - 14)(n - 4) \leq 0$$



$$n \in [4, 14]$$

QUESTION - 9



#Q. Find the number of positive integral values of $3x$, where x satisfy the inequality $(3x - 2)(x - 4) < 0$.

A 10

B 8

C 9 ✓

D none



$$x \in \left(\frac{2}{3}, 4 \right)$$

$$3x \in (2, 12)$$

3, 4, ..., 11



Topics : Special Cases

- INEQUALITY



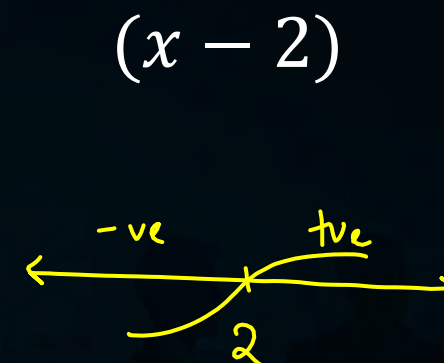
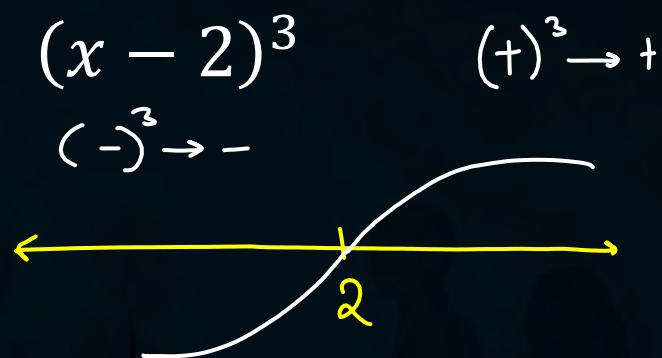
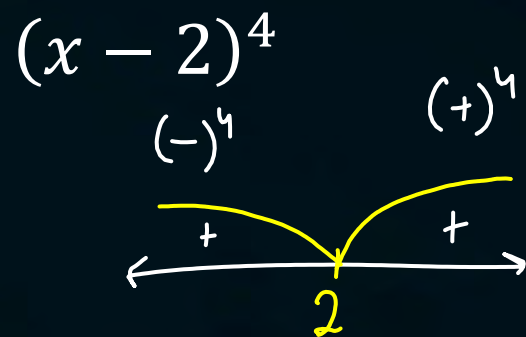
- STEP 1 - Factorize



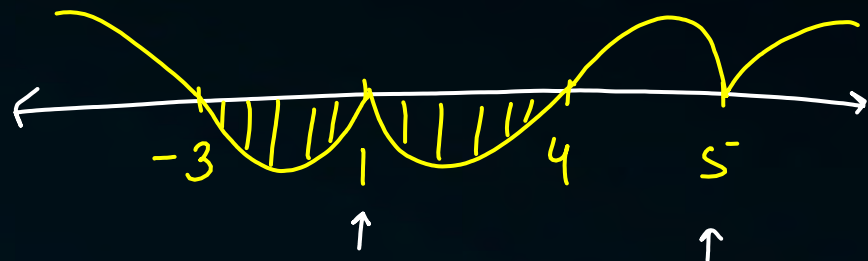
- Critical Cases (FACTORS)



- Sign of Bracket



$$(x - 1)^2(x - 5)^8(x + 3)(x - 4)^3 < 0$$

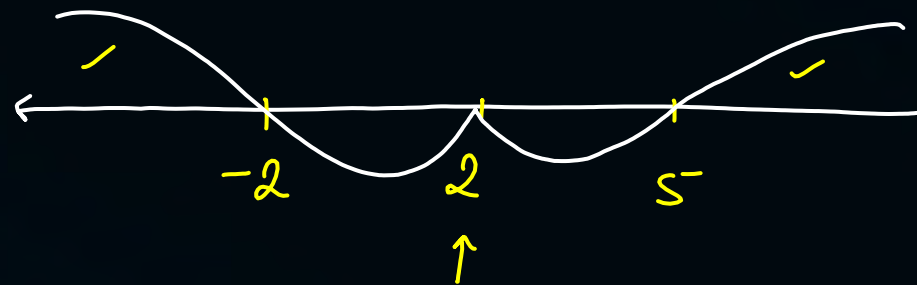


$$\checkmark (-3, 1) \cup (1, 4)$$

$$\times (-3, 4) \longrightarrow \text{hompak} \rightarrow 1$$

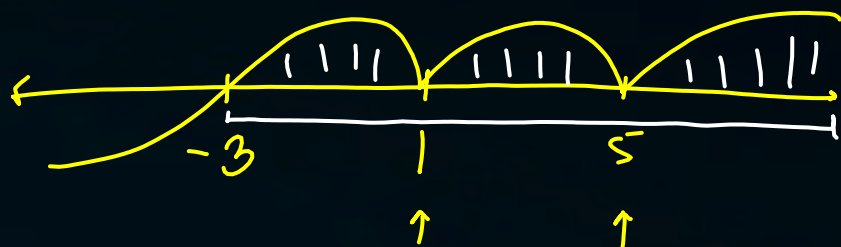
$$\checkmark (-3, 4) - \{1\}$$

$$(x - 2)^2(x - 5)^3(x + 2) > 0$$



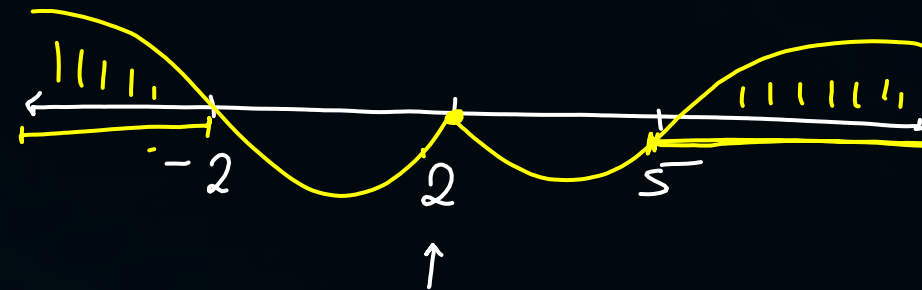
$$\checkmark (-\infty, -2) \cup (5, \infty)$$

$$(x - 1)^2(x - 5)^8(x + 3) \geq 0$$



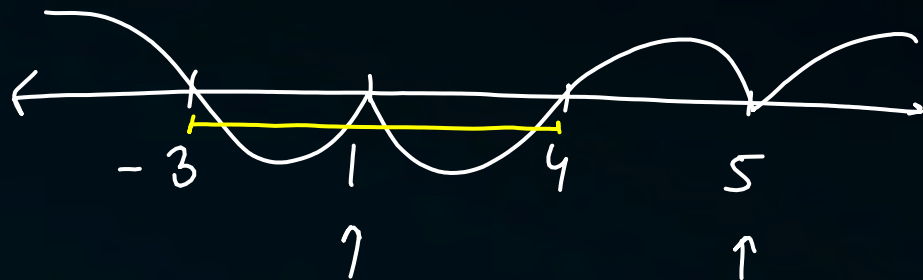
$$[-3, \infty)$$

$$(x - 2)^2(x - 5)^3(x + 2) \geq 0$$



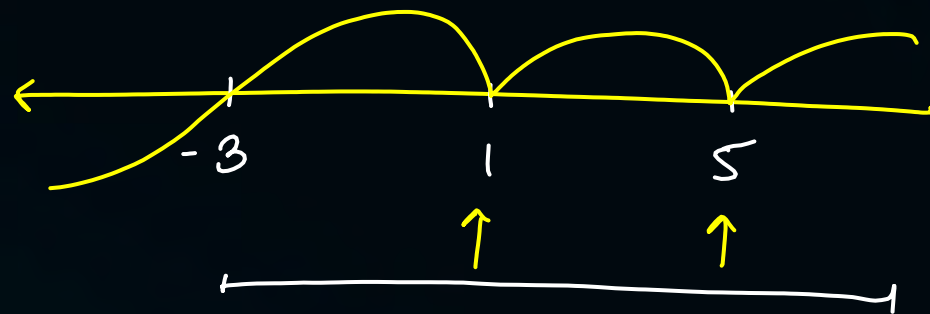
$$(-\infty, -2] \cup [5, \infty) + \{2\}$$

$$(x - 1)^2(x - 5)^8(x + 3)(x - 4)^3 < 0$$



$$(-3, 4) - \{1\}$$

$$(x - 1)^2(x - 5)^8(x + 3) \geq 0$$



$$[-3, \infty)$$



QUESTION - 10



#Q. If $(x - 1)^{2017}(x - 3)^{2019}(x - 5)^{2021}(x - 7)^{2023} < 0$, then how many integrals value can x take.

A

1

B

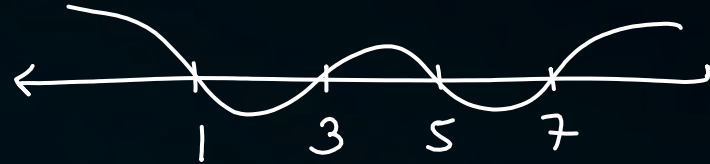
2 ✓

C

3

D

4

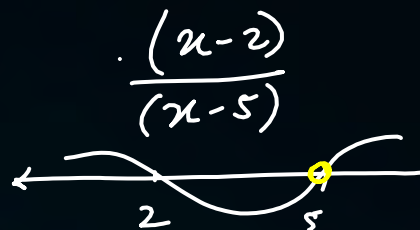


$$\left. \begin{array}{l} x = 2 \\ x = 6 \end{array} \right\} \checkmark$$

$$(+)(+) > 0$$

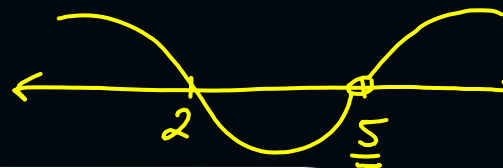
$$\frac{+}{+} > 0$$

$$(x-2)(x-5)$$



$$\frac{-}{-} > 0$$

$$(-)(-) > 0$$



Fractional Inequality

$$(-)(+) < 0$$

$$\frac{-}{+} < 0$$

QUESTION- 11



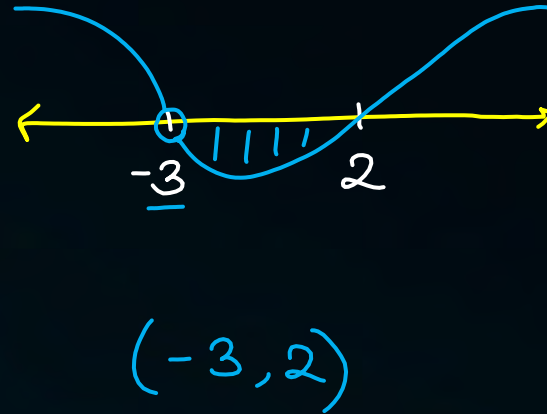
#Q. $\frac{x-2}{x+3} < 0$ find the range of x.

A $x \in (-3, 2)$ //

B $x \in [-3, 2)$

C $x \in (-3, 2]$

D $x \in [-3, 2]$



QUESTION- 12



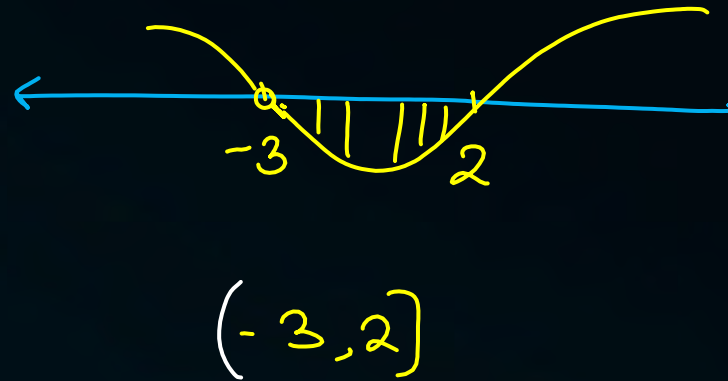
#Q. $\frac{x-2}{x+3} \leq 0$ find the range of x,
less than or equal to 0

A $x \in (-3, 2)$

B $x \in [-3, 2)$

C $x \in (-3, 2]$ ✓

D $x \in [-3, 2]$



QUESTION- 13

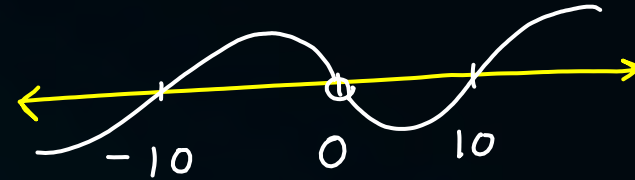


#Q. $\frac{x}{4} \geq \frac{25}{x}$ find the range of x .

$$\frac{x}{4} - \frac{25}{x} \geq 0$$

$$\frac{x^2 - 100}{4x} \geq 0$$

$$\frac{(x-10)(x+10)}{x} \geq 0$$



$$[-10, 0) \cup [10, \infty)$$

QUESTION – 14 (CAT 2024 – SLOT 2)



#Q. All the values of x satisfying the inequality $\frac{1}{x+5} \leq \frac{1}{2x-3}$ are

A $x < -5$ *or* $x > \frac{3}{2}$ ✗

B ✓ $x < -5$ *or* $\frac{3}{2} < x \leq 8$

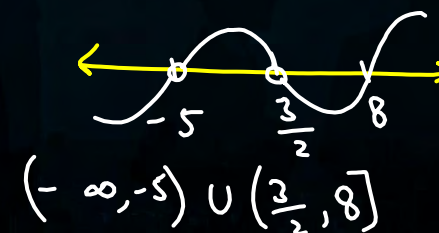
C $-5 < x < \frac{3}{2}$ *or* $\frac{3}{2} < x \leq 8$

D $-5 < x < \frac{3}{2}$ *or* $x > \frac{3}{2}$ ✗

$$\frac{1}{x+5} - \frac{1}{2x-3} \leq 0$$

$$\frac{2x-3 - x-5}{(x+5)(2x-3)} \leq 0$$

$$\frac{(x-8)}{(x+5)(2x-3)} \leq 0$$



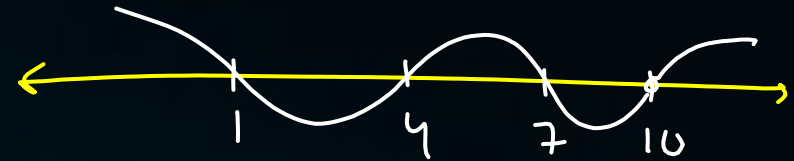
QUESTION - 15



#Q. If $\frac{x^3 - 12x^2 + 39x - 28}{x-10} \geq 0$, Find x .

$$\frac{(x-1)(x-4)(x-7)}{(x-10)} \geq 0$$

- A** $(-\infty, 1) \cup (4, 7) \cup (10, \infty)$
- B** $(-\infty, 1] \cup (4, 7] \cup (10, \infty)$
- C** $(-\infty, -1] \cup [4, 7] \cup [10, \infty)$ ✗
- D** $(-\infty, 1] \cup [4, 7] \cup (10, \infty)$





$$x^2 + x + 1 \longrightarrow \text{Always } +ve$$

$$a > 0$$

$$D < 0$$



$$x^2 + x + 1 > 0$$

$$x \in \mathbb{R} \quad x \in (-\infty, \infty)$$

$$x^2 + x + 1 < 0$$

$$x \in \phi$$

No sol.

$$-x^2 - x - 1 \longrightarrow \text{Always } -ve$$

$$a < 0$$

$$D < 0$$



$$-x^2 - x - 1 < 0 \quad | \quad -x^2 - x - 1 > 0$$

$$x \in \mathbb{R}$$

$$x \in \phi$$

QUESTION – 16



#Q. All the values of x satisfying the inequality $\frac{(x^2 + x + 1)}{5} > 0$

$$\frac{(+)}{(+)}$$

✓ $x \in \mathbb{R}$

QUESTION – 17



#Q. All the values of x satisfying the inequality $\frac{-(x^2 + x + 1)}{5} > 0$

$$x \in \phi$$

#Q. All the values of x satisfying the inequality $(x^2 + x + 1)(x^2 + 5x + 6) \geq 0$

$$(x^2 + 5x + 6) \geq 0$$

$$(x+2)(x+3) \geq 0$$



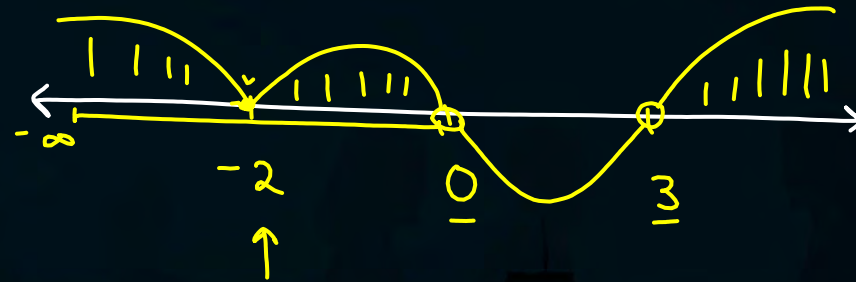
$$(-\infty, -3] \cup [-2, \infty)$$

QUESTION – 19



#Q. All the values of x satisfying the inequality $\frac{(x^2 + x + 1)(x + 2)^4}{x^2 - 3x} \geq 0$

$$\frac{(x + 2)^4}{x(x - 3)} \geq 0$$



$$(-\infty, 0) \cup (3, \infty)$$

#Q. If the values of x satisfying the inequality $x^2 + mx + n < 0$ are $(-2, 5)$, then find the value of $m + n$.

- A** 13
- B** 7
- C** -13
- D** none

HW

→ MBA WALLAH ALGEBRA foundation
PLAYLIST

→ functions ✓

→ MBA WALLAH functions ESSENTIALS



THANK
You

